

SECR2043 OPERATING SYSTEMS

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Section : 02

Marks

This lab assessment is designed to test your understanding and skills on some basic concepts and tools related to operating system. Please follow the instructions carefully and submit your answers in this word document and rename the file as **os-lab-assessment01-studentname-matricno.docx**.

ACTIVITY 1 : BASIC LINUX COMMANDS

In this activity, you will practice some basic Linux commands that are useful for operating system tasks. For each question, write down the command you used and the output you got in your answer file. If the command does not produce any output, write "**No output**" instead.

1.	Display your current working directory.	
	Command	Output
	pwd	evelyn@secr2043:~\$ pwd /home/evelyn
2.	List all the files and directories in your home directory, including hidden ones.	
	Command	Output
	ls -a ~	evelyn@secr2043:~\$ ls -a ~bash_logout .bashrc .profile
3.	Create a new directory named "os_lab" in your home directory.	
	Command	Output
	mkdir os_lab	evelyn@secr2043:~\$ mkdir os_lab evelyn@secr2043:~\$ ls -a ~bash_logout .bashrc .profile os_lab
4.	Change your current working directory to "os_lab".	
	Command	Output
	cd os_lab	evelyn@secr2043:~\$ pwd /home/evelyn evelyn@secr2043:~\$ cd os_lab evelyn@secr2043:~/os_lab\$ pwd /home/evelyn/os_lab
5.	Create a new file named "hello.txt" in "os_lab" and write "Hello, world!" in it.	
	Command	Output

	nano hello.txt	evelyn@secr2043:~/os_lab\$ ls evelyn@secr2043:~/os_lab\$ nano hello.txt evelyn@secr2043:~/os_lab\$ ls hello.txt
6.	Display the content of "hello.txt".	
	Command	Output
	cat hello.txt	evelyn@secr2043:~/os_lab\$ cat hello.txt Hello, world!
7.	Copy "hello.txt" to another file named "hello_copy.txt" in the same directory.	
	Command	Output
	cp hello.txt hello_copy.txt	evelyn@secr2043:~/os_lab\$ ls hello.txt evelyn@secr2043:~/os_lab\$ cp hello.txt hello_copy.txt evelyn@secr2043:~/os_lab\$ ls hello.txt hello_copy.txt
8.	Rename "hello_copy.txt" to "hello_bak.txt".	
	Command	Output
	mv hello_copy.txt hello_bak.txt	evelyn@secr2043:~/os_lab\$ mv hello_copy.txt hello_bak.txt evelyn@secr2043:~/os_lab\$ ls hello.txt hello_bak.txt
9.	Delete "hello.txt".	
	Command	Output
	rm hello.txt	evelyn@secr2043:~/os_lab\$ ls hello.txt hello_bak.txt evelyn@secr2043:~/os_lab\$ rm hello.txt evelyn@secr2043:~/os_lab\$ ls hello_bak.txt
10.	Display your home directory with additional information, together with subdirectories contents.	
	Command	Output
	ls -R /home	evelyn@secr2043:~/os_lab\$ ls -R /home /home: evelyn farkhana mahyuddin student /home/evelyn: os_lab /home/evelyn/os_lab: hello_bak.txt
	Total Mark:	

ACTIVITY 2 : Text File Manipulation

In this activity, you will practice some commands to create and manipulate text files using echo, cat and touch.

For each question, write down the command and the output (if any) in the answer sheet.

1.	Create an empty file named "hello.txt" in the current working directory.	
	Command	Output
	touch hello.txt	<pre>evelyn@secr2043:~\$ ls os_lab evelyn@secr2043:~\$ touch hello.txt evelyn@secr2043:~\$ ls hello.txt os_lab</pre>
2.	Write the text "Hello World" to "hello.txt" using echo.	
	Command	Output
	echo "Hello World" > hello.txt	<pre>evelyn@secr2043:~\$ echo "Hello World" > hello.txt</pre>
3.	Append the text "This is a test" to "hello.txt" using echo.	
	Command	Output
	echo "This is a test" >> hello.txt	<pre>evelyn@secr2043:~\$ echo "This is a test" >> hello.txt</pre>
4.	Display the contents of "hello.txt" using cat.	
	Command	Output
	cat hello.txt	<pre>evelyn@secr2043:~\$ cat hello.txt Hello World This is a test</pre>
5.	Create another file named "bye.txt" with the text "Goodbye World" using echo.	
	Command	Output
	touch bye.txt echo "Goodbye World" > bye.txt	<pre>evelyn@secr2043:~\$ touch bye.txt evelyn@secr2043:~\$ echo "Goodbye World" > bye.txt</pre>
6.	Concatenate the contents of "hello.txt" and "bye.txt" and display them using cat.	
	Command	Output
	cat hello.txt bye.txt	<pre>evelyn@secr2043:~\$ cat hello.txt bye.txt Hello World This is a test Goodbye World</pre>
7.	Concatenate the contents of "hello.txt" and "bye.txt" and write them to a new file named "greeting.txt" using cat.	
	Command	Output

	cat hello.txt bye.txt > greeting.txt	evelyn@secr2043:~\$ cat hello.txt bye.txt > greeting.txt
8.	Display the contents of "greeting.txt" using cat.	
	Command	Output
	cat greeting.txt	evelyn@secr2043:~\$ cat greeting.txt Hello World This is a test Goodbye World
9.	Edit the file "greeting.txt" using Nano and change the word "World" to your name in both lines.	
	Command	Output
	nano greeting.txt	evelyn@secr2043:~\$ nano greeting.txt GNU nano 7.2 Hello Evelyn This is a test Goodbye Evelyn
10.	Display the modified contents of "greeting.txt" using cat.	
	Command	Output
	cat greeting.txt	evelyn@secr2043:~\$ cat greeting.txt Hello Evelyn This is a test Goodbye Evelyn
	Total Mark:	

ACTIVITY 3 : WRITE, COMPILE AND RUN C PROGRAM

In this activity, you will practice how to write, compile and run a simple C program using gcc. For each question, write down the command and the output (if any) in the answer sheet.

1.	Write a C program that prints "Hello World" to the standard output using Nano and save it as "hello.c".	
	Command	Output
	nano hello.c	<pre>evelyn@secr2043:~\$ nano hello.c GNU nano 7.2 #include <stdio.h> int main() { printf("Hello, World!\n"); return 0; }</pre>
2.	Compile the program "hello.c" using gcc and generate an executable file named "hello".	
	Command	Output
	gcc hello.c -o hello	<pre>evelyn@secr2043:~\$ gcc hello.c -o hello</pre>
3.	Run the executable file "hello" and display its output.	
	Command	Output
	./hello	<pre>evelyn@secr2043:~\$./hello Hello, World!</pre>
4.	Write a C program that takes two integers as command line arguments and prints their sum to the standard output using Nano and save it as "sum.c".	
	Command	Output
	nano sum.c	<pre>evelyn@secr2043:~\$ nano sum.c GNU nano 7.2 # include <stdio.h> int main(){ int x,y; printf("Enter first number: "); scanf("%d", &x); printf("Enter second number: "); scanf("%d", &y); int sum=x+y; printf("The sum of %d and %d is: %d\n", x, y, sum); return 0; }</pre>
5.	Compile the program "sum.c" using gcc and generate an executable file named "sum".	
	Command	Output
	gcc sum.c -o sum	<pre>evelyn@secr2043:~\$ gcc sum.c -o sum</pre>

6.	Run the executable file "sum" with 10 and 20 as arguments and display its output.	
	Command	Output
	./sum	evelyn@secr2043:~\$./sum Enter first number: 10 Enter second number: 20 The sum of 10 and 20 is: 30
7.	Run the executable file "sum" with -5 and 15 as arguments and display its output.	
	Command	Output
	./sum	evelyn@secr2043:~\$./sum Enter first number: -5 Enter second number: 15 The sum of -5 and 15 is: 10
8.	Write a C program that reads a line of text from the standard input and prints it to the standard output using Nano and save it as "echo.c".	
	Command	Output
	nano echo.c	evelyn@secr2043:~\$ nano echo.c GNU nano 7.2 #include <stdio.h> int main(){ char echo[50]; printf("Enter the text: "); fgets(echo,50,stdin); printf("The test you input is: %s\n",echo); return 0; }
9.	Compile the program "echo.c" using gcc and generate an executable file named "echo".	
	Command	Output
	gcc echo.c -o echo	evelyn@secr2043:~\$ gcc echo.c -o echo
10.	Run the executable file "echo", type "Hello World" as input and display its output.	
	Command	Output
	./echo	evelyn@secr2043:~\$./echo Enter the text: Hello World The test you input is: Hello World
	Total Mark:	